

CS23532-COMPUTER NETWORKS

EXPERIMENT-14

AIM: - Write a code using RAW sockets to implement packet sniffing.

ALGORITHM

packet_sniffer.py

```
import socket

HOST = socket.gethostbyname(socket.gethostname())

sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)

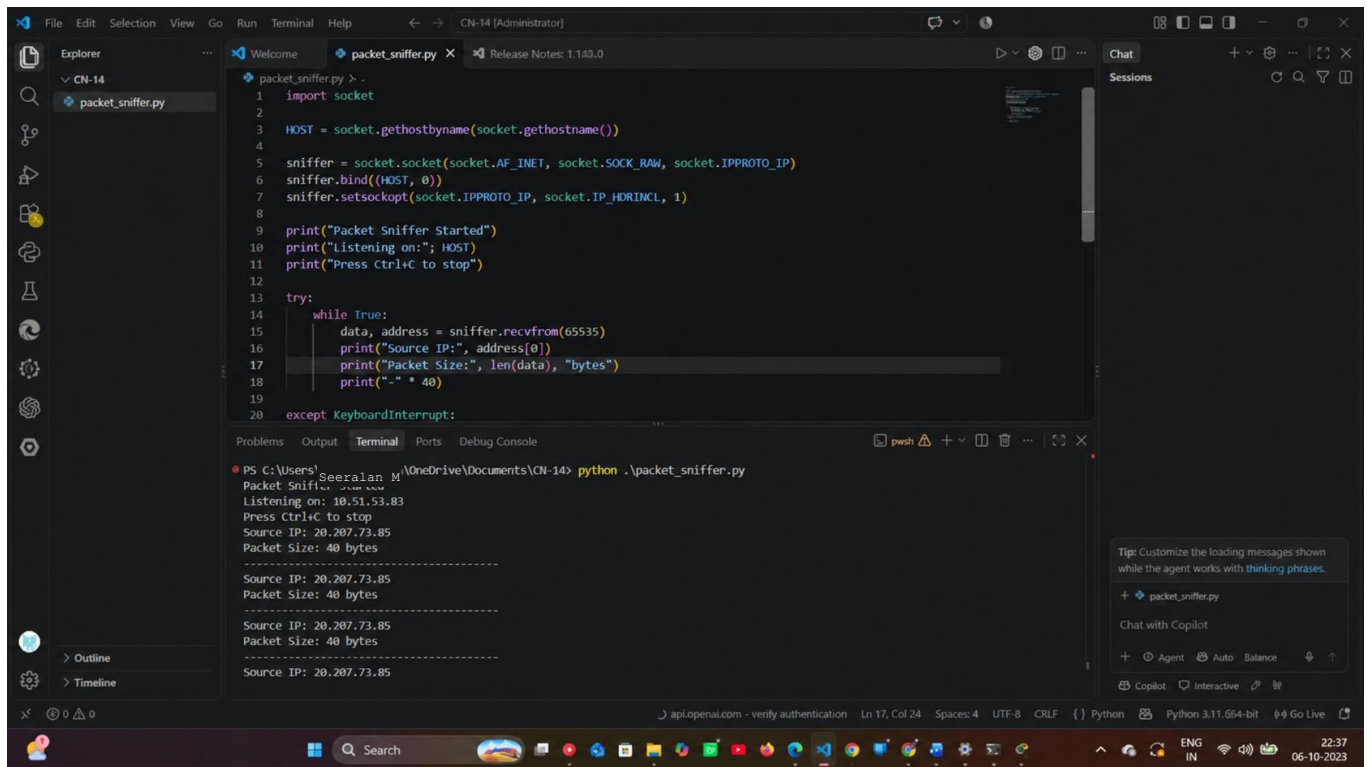
sniffer.bind((HOST, 0)) sniffer.setsockopt(socket.IPPROTO_IP,
socket.IP_HDRINCL, 1)

print("Packet Sniffer Started")

print("Listening on:", HOST) print("Press
Ctrl+C to stop")

try:
    while True:
        data, address = sniffer.recvfrom(65535)
        print("Source IP:", address[0])    print("Packet
Size:", len(data), "bytes")    print("-" * 40)
except KeyboardInterrupt:
    print("\nPacket Sniffer Stopped")

finally:
    sniffer.close()
```



```
File Edit Selection View Go Run Terminal Help
CN-14 [Administrator]

Welcome
packet_sniffer.py x Release Notes: 1.143.0

packet_sniffer.py > -
1 import socket
2
3 HOST = socket.gethostname(socket.gethostname())
4
5 sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
6 sniffer.bind((HOST, 0))
7 sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
8
9 print("Packet Sniffer Started")
10 print("Listening on: ", HOST)
11 print("Press Ctrl+C to stop")
12
13 try:
14     while True:
15         data, address = sniffer.recvfrom(65535)
16         print("Source IP:", address[0])
17         print("Packet Size:", len(data), "bytes")
18         print("-" * 40)
19
20 except KeyboardInterrupt:
```

Problems Output Terminal Ports Debug Console

```
PS C:\Users\Seeralan M\OneDrive\Documents\CN-14> python .\packet_sniffer.py
Packet Sniffer Started
Listening on: 10.51.53.83
Press Ctrl+C to stop
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
```

Tip: Customize the loading messages shown while the agent works with *thinking phrases*.

+ packet_sniffer.py

Chat with Copilot

+ Agent Auto Balance

Copilot Interactive

api.openai.com - verify authentication Ln 17, Col 24 Spaces: 4 UTF-8 CRLF Python Python 3.11.654-bit Go Live

22:37 06-10-2023

